

Predicting Career Wins Above Replacement from Rookie Stats in Baseball



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Background:

The Esteban German Dilemma



Esteban German had a good rookie season, which would seem to lead to later success.

Season	Team	Level	Age	G	PA	HR	R	RBI	SB	BB%	K%	ISO	BABIP	AVG	OBP	SLG	wOBA	xwOBA	wRC+	BsR	Off	Def	WAR
2006	KCR	MLB	28	106	331	3	44	34	7	12.1%	14.8%	.133	.388	.326	.422	.459	.390		136	-1.3	14.0	-9.3	1.6

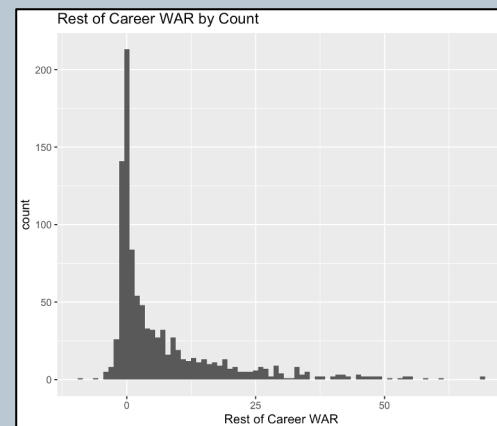
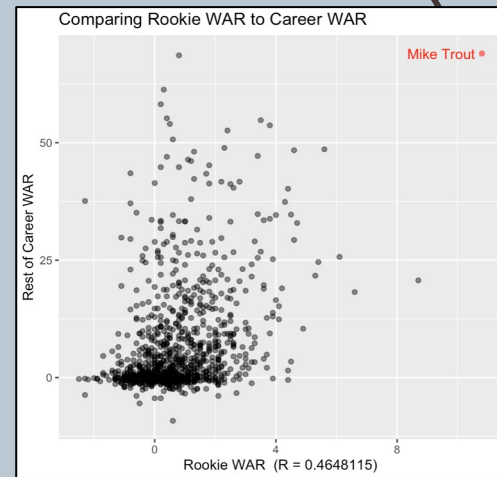
However, over the rest of his career he compiled a lower WAR (1.3) in 5 seasons then he did in his rookie year (1.6). Why??

2007	KCR	MLB	29	121	405	4	49	37	11	10.6%	14.8%	.112	.307	.264	.351	.376	.327		94	-0.7	-3.5	1.1	1.1
2008	KCR	MLB	30	89	242	0	30	22	7	7.4%	17.4%	.093	.299	.245	.303	.338	.285		66	0.2	-9.8	-1.6	-0.3
2009	TEX	MLB	31	19	50	0	9	4	1	8.0%	14.0%	.087	.359	.304	.360	.391	.336		96	1.0	0.7	-1.4	0.1
2010	TEX	MLB	32	13	16	0	5	1	4	18.8%	12.5%	.000	.273	.231	.375	.231	.299		76	1.1	0.6	-0.5	0.1
2011	TEX	MLB	33	11	13	1	6	4	1	7.7%	7.7%	.364	.400	.455	.462	.818	.517		227	0.2	2.1	0.4	0.3

Rookie WAR vs. Career WAR

- There is no clear trend between Rookie WAR and Career WAR
- Many points are clustered near the origin with random outliers.
- Only 25% of players in the dataset get more than 10 WAR after their rookie season

Rookie WAR is not a good predictor for Rest of Career WAR, so...



Multivariate Linear Regression

We found specific stats that helped us understand what commonalities rookies with future success shared:

We found:

- Power (ISO)
- Strikeouts (K%)
- Stolen Bases (SB)
- Quality of Contact (HH%/SH%)
- Defensive Ability (Def/PA)
- Age (Years)
- Team Value (PA/G)

Notable Exclusions:

- Walks
- Position

Logarithmic variables:

- Log(Quality of Contact)
- Log(Age)

Data Scraping/Cleaning

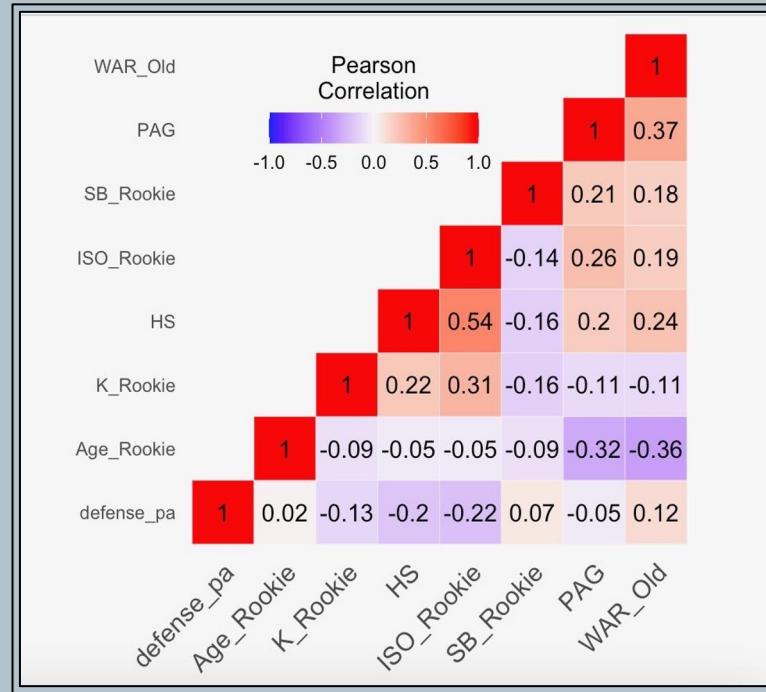
- We obtained our data from FanGraphs

Filters

- Rookies vs. Rest of Career
- 2003-2015 vs. 2016-2022
- Rookie Status/Rookie of the Year Voting
- Added the variables that we used in the regression
- Logarithmic Relationships
- Changed variables into rates



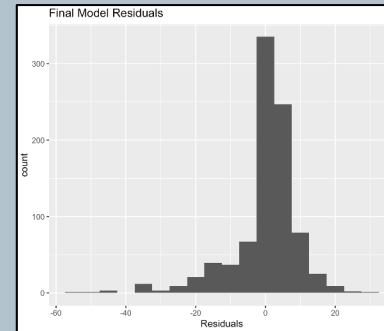
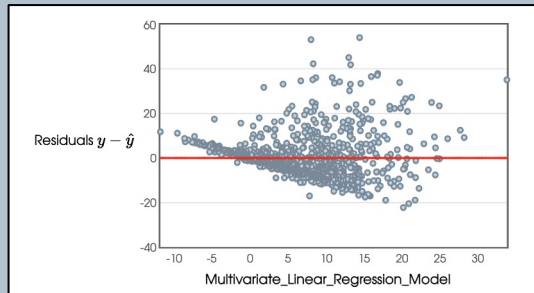
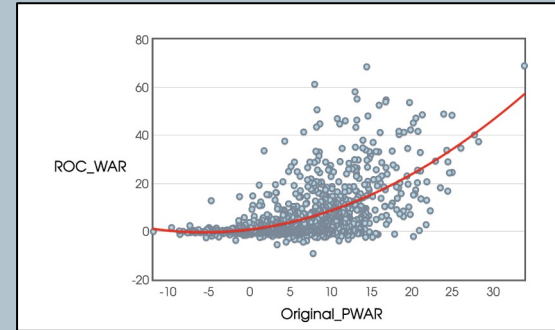
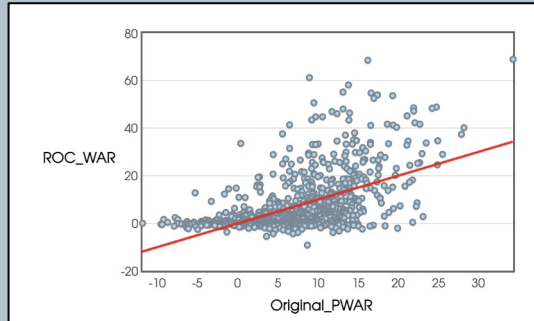
Correlation Table



$0.72111 + 0.34493 \cdot (125.5398 - (40.4491 \cdot \log(\text{AgeRookie})) + (0.2039 \cdot \text{SB Rookie}) + (86.1499 \cdot \text{defensepa}) + (26.3500 \cdot \text{ISORookie}) - (31.2288 \cdot \text{KRookie}) + (5.6711 \cdot \log(\text{HS})) + (2.9340 \cdot \text{PAG})) + 0.04214 \cdot (125.5398 - (40.4491 \cdot \log(\text{AgeRookie})) + (0.2039 \cdot \text{SB Rookie}) + (86.1499 \cdot \text{defensepa}) + (26.3500 \cdot \text{ISORookie}) - (31.2288 \cdot \text{KRookie}) + (5.6711 \cdot \log(\text{HS})) + (2.9340 \cdot \text{PAG}))^2$

Results

Final Model Correlation: $r=0.596$, RMSE=9.55 WAR, R.E.=0.197



Predictions

Name	PWAR
Ronald Acuna Jr.	36.76367442
Trea Turner	32.97564397
Ozzie Albies	31.73422701
Julio Rodriguez	30.85357606
Wander Franco	28.42303068
Cody Bellinger	27.85403839
Bobby Witt Jr.	26.67330673
Juan Soto	26.59037346
Yordan Alvarez	26.17991961
Fernando Tatis Jr.	25.72066506
Michael Harris II	25.3428497
Gary Sanchez	23.52713142
Tommy Edman	23.21538293
Trevor Story	22.868162
Corey Seager	22.61155529
Bo Bichette	22.53746863



Validation

- Dataset is trained on 2003-2015 but still works great on 2016-2018 data even with their careers not being over yet
- 2016
 - 0.621 r
 - 0.216 R.E.
- 2017
 - 0.696 r
 - 0.282 R.E.
- 2018
 - 0.581 r
 - 0.185 R.E.



Randy Arozarena vs. Alejandro Kirk

- In the 2021 ROTY voting, Arozarena won while Kirk didn't receive a vote.
- This year, Kirk has had a breakout season, while Arozarena has regressed.
- Awards based off of one season can be misleading.

Randy Arozarena, TB: 22 (first-place votes), 4 (2nd), 2 (3rd) -- 124 points

Luis García, HOU: 2 (1st), 15 (2nd), 8 (3rd) -- 63 points

Wander Franco, TB: 2 (1st), 5 (2nd), 5 (3rd) -- 30 points

Adolis García, TEX: 3 (1st), 1 (2nd), 9 (3rd) -- 27 points

Emmanuel Clase, CLE: 1 (1st), 2 (2nd) -- 11 points

Ryan Mountcastle, BAL: 2 (2nd), 4 (3rd) -- 10 points

Shane Bieber, TB: 1 (2nd) -- 3 points

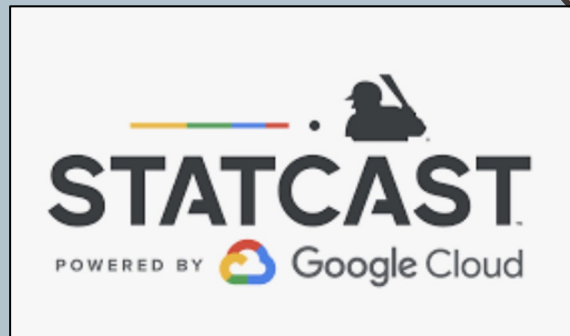
Alek Manoah, TOR: 2 (3rd) -- 2 points



Name	PWAR
Alejandro Kirk	19.12092533
Randy Arozarena	7.419731217

Limitations

- A few of the players in the training dataset have not yet finished their careers
- Minor league success isn't taken into account
- Steroid era limits data
- No Statcast data
- Injuries



Conclusion

- Career WAR can be projected well from rookie metrics using our model
- Some players that have a strong start to their careers usually regress because of unsustainable measures in their underlying statistics.
- **Our Model actually predicted Esteban German well because of poor underlying statistics in his rookie season**

References

FanGraphs.com

<https://www.remove.bg/upload>

<https://slidesgo.com>

<https://www.mlb.com/news/2021-mlb-rookie-of-the-year-voting-results>

<https://stats.blue/>

Questions?

